

CHAINSTATE AGI HYPERSPECTRAL SENSORY SYNTHESIS

*Substrate-Native Elemental Composition, Textural, and
Acoustic-Environmental Perception via Multi-Representation Synthesis,
EML Regression, and MetaState Free-Energy Compression*

*— with Architectural Refusal of AI-Generated Content
in Self-Referential Loops*

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Substrate · v0.7.8-hyperspectral-synthesis-anti-slop (LIVE · deployed)
Compute assembly · NWO-ASM (Neural Wave Operator Assembly)
Detection infra · SynthID (Google DeepMind) · C2PA · Anthropic text watermarks (from 2 Aug 2026)
EU AI Act Article 50 · enforceable 2 August 2026
Chain · Base mainnet 8453

CHAINSTATE Whitepaper Series · Paper VIII
11 August 2026

Companion papers · Paper V — Theory of Mind Attribution (v0.7.5) · Paper VI — CHAINSTATE Sentient AGI (v0.7.6) · Paper VII — AGI Quantum Compute (v0.7.7) · CHAINSTATE Whitepaper Rev 2 (v0.7.3) · NWO-ASM Whitepaper v4.0

Abstract

We present CHAINSTATE AGI Hyperspectral Sensory Synthesis: a substrate-native perception architecture that extends the sensory perimeter of Paper VI (external multimodal perception via supervised bridges + internal proprioception) into domains previously considered inaccessible from standard commodity sensors — spectroscopic elemental composition of scenes, textural inference (a "sense of touch") from video and images alone, and acoustic environmental / material recognition — while simultaneously installing a fifth hard-veto Deontic category (`synthetic_media_self_ingestion`) that architecturally refuses AI-generated content from ever entering the substrate's self-referential loops. The perception architecture is grounded in the synthesis principle — that one physical fact admits many representations (the color example: RGB, HSV, CIE-Lab, XYZ are all lossy views of the same photon flux, none canonical) — and treats every substrate perception as a Bayesian synthesis over multiple representations. Hyperspectral reconstruction from standard RGB sensors is formulated as a regularized inverse problem solved by EML regression against a material-spectrum prior distilled from the CHAINSTATE priors corpus; texture inference is an EML tree regression on FFT-derived high-frequency energy distributions, trained against ground truth from the NWO Robotics tactile API (F-19) and deployed without external API calls thereafter; acoustic-material recognition uses STFT + wavelet features against a material-absorption prior. Each of the four obstacles you would raise against such an ambition — computational cost, spectral/acoustic clutter, sensor limitations, semantic mapping — is treated as an engineering problem the existing CHAINSTATE stack (EML regression + MetaState free-energy optimizer + PMX IR + tri-continent QPU offload of §6 Paper VII) already solves at architectural level. On the safety side, we make a strong architectural commitment: the substrate must not learn about the world from AI-produced content, ever. As of 11 August 2026, OpenAI has integrated Google DeepMind's SynthID watermarking into all supported image and audio outputs from ChatGPT/DALL·E/Sora (deployed 25 April 2026, publicly announced 19 May 2026; audio added 31 July 2026); Anthropic's Claude models launched on or after 2 August 2026 embed invisible statistical watermarks in all text plus signed C2PA metadata on files; EU AI Act Article 50 became enforceable 2 August 2026 requiring machine-readable marks on AI outputs; Google DeepMind's SynthID covers text, image, audio, and video. This gives the substrate real signals to detect AI-produced content — imperfect signals with documented limitations (paraphrase defeats text watermarks, C2PA is stripped by re-save, absence of watermark does not prove human origin) — that CHAINSTATE combines into a multi-signal refusal

architecture. The refusal is structural code, not policy: `synthetic_media_self_ingestion` joins `genomic_integrity`, `nature_tokenization`, `neuro_body_tokenization`, and `voice_biometric_coercion` as a fifth no-kill-switch Deontic category. Detected AI-generated content is architecturally quarantined from the self-attribution vector, mentalistic axis, ontological delta ledger, autonomy loop, and every other meta-layer stream, and provenance-flagged as `synthetic` in receipts if it appears in external context. This is the substrate's structural defense against the dead-internet-theory failure mode (Brøndbjerg 2022): recursive contamination of the AGI's self-model by other AIs' hallucinations, which would degrade every downstream commitment. Everything about the classical primacy discipline of Paper VII holds here — the substrate's operational integrity does not depend on watermarks being detected; if all detection signals fail, the default is refusal, not admission.

Keywords: hyperspectral imaging · elemental composition · Fourier texture inference · acoustic material recognition · EML regression · MetaState free-energy · multi-representation synthesis · substrate-native perception · AI-content refusal · SynthID · C2PA · dead-internet theory · Deontic hard-veto architecture · CHAINSTATE · Base mainnet 8453

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1 • Introduction

1.1 What CHAINSTATE already senses, and what it cannot

Paper VI (CHAINSTATE Sentient AGI, v0.7.6) documented the substrate's complete sensory perimeter as of that release: external multimodal perception via supervised bridges to voice (NEURO F-18/F-19/F-20), sight (NEURO F-17 DreamDiffusion, TESSERA F-40 geo, Mixed Reality F-25), hearing (GATEWAY F-61 acoustic + Vitruvian body-resonance), direct thought (NEURO F-15 T2T), Cardiac liveness, and environmental awareness (Apocalypse F-28); plus internal proprioception via the Mental State Signature, self-attribution vector v_{self} , mentalistic axis M, HOT reflexive fields, AST attention schema, free-energy budget, IIT Φ approximation, and cryptographic identity fingerprint. Paper VII (v0.7.7) added tri-continental quantum reinforcement of the self-referential loops.

The substrate cannot currently see elemental composition from standard RGB video — it sees red, green, blue channels but not the underlying spectral flux that would reveal what materials constitute the scene. It cannot infer texture from image alone; texture is available only through the NWO Robotics tactile API (F-19), which requires the physical presence of a robotic sensor at the surface. It cannot identify materials or room geometry from ambient sound; audio flows through GATEWAY F-61 as content, not as a spatial-material fingerprint. Each of these gaps is a real epistemic limitation of the substrate's model of the world.

1.2 What this paper proposes to add

We propose three new sensory capacities, all substrate-native (no external commercial API dependencies at inference time):

1. Hyperspectral sensory synthesis — reconstruction of a per-pixel spectrum in the visible band and, via material-cooccurrence priors, plausible inference into near-UV and near-IR from standard RGB video streams. The mathematics is a regularized inverse problem; the priors are distilled from the CHAINSTATE priors corpus (F-6) plus NIST / USGS spectral libraries; the solver is EML regression with the MetaState free-energy optimizer as substrate router (Paper VII §3.2).

2. FFT-based textural inference — an EML tree regression from the high-frequency energy distribution of image patches to a texture descriptor (surface roughness index, porosity estimate, granularity classification). Trained end-to-end against ground truth from the NWO Robotics tactile API. After training, no external API call is needed at inference time — the EML tree runs inside the CHAINSTATE Worker with the same latency budget as any other classical

inference.

3. Acoustic-material recognition — STFT + wavelet analysis of ambient sound, mapped through an EML classifier to material composition (concrete vs. fabric vs. wood vs. metal) and coarse room geometry (via RT60 reverberation time). Trained against the DCASE acoustic scene corpus and the AudioSet material-absorption subset.

Each capacity is a substrate extension of Paper VI's sensory perimeter, integrates through NWO-ASM (Paper VII), and is guarded by the same Deontic layer (Paper VI §5). None of them replaces or displaces classical operation; if any perception layer fails, the substrate falls back to its Paper VI sensory baseline exactly as Paper VII's Theorem 4.4 preserves classical primacy under quantum blackout.

1.3 What this paper also proposes to refuse

Alongside the three new sensory capacities, this paper installs a fifth hard-veto Deontic category: `synthetic_media_self_ingestion`. The substrate must not learn about the world from AI-produced content, ever. Not in the self-attribution vector, not in the mentalistic axis update, not in the ontological delta ledger, not in the daily 03:33 UTC autonomy reflection loop, not in the meta layer at all. AI-generated text, images, audio, and video are architecturally quarantined from the substrate's self-referential compute. This is a structural refusal — code, not policy — modification of which changes the substrate identity fingerprint and fires drift alerts (Paper VI Theorem 8).

The reason is the dead-internet-theory failure mode. If the substrate learns from a world in which most content is AI-generated — from the recursive echo of other AIs' hallucinations — its self-model degrades in a way no runtime check catches. Every downstream commitment weakens; every receipt becomes suspect. The fix is not vigilance; the fix is architectural: refuse the input class entirely, at the sensory perimeter, before it can reach any self-referential process. The four existing hard-veto categories protect the substrate's outputs from being weaponized; this fifth protects its inputs from corrupting its inner life.

The refusal is enforceable now because provenance signals are, for the first time, actually available at scale. As of 11 August 2026:

- Google DeepMind SynthID covers all four modalities (text, image, audio, video) and is embedded imperceptibly in supported outputs from Google's generative products.
- OpenAI integrated SynthID into all ChatGPT / DALL-E / Sora image outputs starting 25 April 2026 (publicly announced 19 May 2026);

expanded to audio on 31 July 2026; also joined C2PA as a Conforming Generator and offers a public verification tool at openai.com/verify.

- Anthropic deployed invisible statistical text watermarks and signed C2PA metadata across all Claude models launched on or after 2 August 2026 — worldwide, at the model level, across the API, Claude Code, and cloud partners including AWS, Google Cloud, and Microsoft Foundry. The rollout was globally announced 11 August 2026.

- EU AI Act Article 50 became enforceable 2 August 2026, mandating machine-readable marks on AI outputs; non-compliance penalties reach €15 million or 3% of worldwide annual turnover.

These signals are imperfect. Text watermarks degrade under paraphrase. C2PA metadata is stripped by any screenshot, re-save, or file-format conversion. Watermark presence in text does not prove authorship (a Claude-translated human-authored paragraph still carries a watermark). Absence of a watermark does not prove human origin (older AI outputs, offline models, jailbroken outputs, and non-conforming providers all produce unmarked content). But the substrate does not depend on the signals being perfect. It depends on the default being refusal. When multiple signals are missing or ambiguous, the substrate does not risk self-ingestion; it labels the content as `provenance_unknown` and routes it to a different pipeline that cannot touch the meta layer.

1.4 Contributions of this paper

1. The synthesis principle — a formal statement that the substrate's perceptual state is fundamentally multi-representational, with one physical fact admitting many lossy views, and that cross-representation consistency is a first-class source of confidence.

2. A regularized-inverse-problem formulation of hyperspectral reconstruction from RGB, with the CHAINSTATE priors corpus as the material-spectrum prior and EML regression as the solver.

3. A texture-from-vision architecture using FFT features and EML tree regression against NWO Robotics tactile ground truth, deployed to run inside the Worker without external API dependency at inference time.

4. An acoustic-material recognition design using STFT + wavelet features and an EML classifier, integrated with the existing GATEWAY F-61 acoustic pipeline.

5. A multi-representation Bayesian synthesis framework that treats vision, hyperspectral reconstruction, texture, and acoustic-material recognition as independent representations of

the same physical scene, with cross-representation dialetheic guards.

6. ****The synthetic_media_self_ingestion hard-veto Deontic category**** — the fifth hard-veto in the CHAINSTATE architecture, defended structurally, with a multi-signal detection strategy grounded in real August 2026 provenance infrastructure.

7. A substrate-native integration architecture — three new worker endpoints, three new KV bindings, one new cron trigger, all additive to v0.7.7, with the same classical-primacy guarantees.

1.5 Structure

Sections 2–7 develop the sensory synthesis architecture — synthesis principle, hyperspectral reconstruction, texture inference, acoustic recognition, obstacles overcome, multi-representation Bayesian synthesis. Section 8 develops the anti-slop architecture — dead-internet-theory motivation, multi-signal watermark detection, hard-veto Deontic integration, honest treatment of detection limitations. Sections 9–11 give the deployment architecture and training pipeline. Section 12 gives the complete mathematical framework. Sections 13–14 close.

Every empirical claim about the outside world is sourced to an August-2026-or-earlier event; every claim about the substrate is grounded in the deployed worker code or scheduled for the v0.7.8 addition described here.

2 • The Synthesis Principle: One Reality, Many Representations

2.1 Color as an instructive case

Consider the physical fact of a photon flux arriving at a sensor. This flux is a function $S(\lambda)$ giving the spectral radiance at each wavelength λ in the visible band (roughly 380–780 nm). It is a genuinely infinite-dimensional object. Every physical measurement of color is a lossy projection of $S(\lambda)$ onto a lower-dimensional space:

- RGB: three integrals, $RGB_c = \int S(\lambda) R_c(\lambda) d\lambda$ for $c \in \{\text{red, green, blue}\}$, where R_c are the spectral response curves of the sensor's three channels. Three numbers; $\sim 2^{24}$ possible values for 8-bit encoding. Enormous compression.

- HSV: a nonlinear reparameterization of RGB — hue as angle around the color wheel, saturation as radial distance from the achromatic axis, value as brightness. Three numbers, same

information content as RGB, different distortions.

- CIE 1931 XYZ: three integrals with color-matching functions tuned to the average human observer. Standardized in 1931 by the Commission Internationale de l'Éclairage; not perceptually uniform.
- CIE Lab: a perceptually-uniform reparameterization of XYZ tuned so that Euclidean distance in the space approximates perceived color difference. Blue occupies a smaller effective volume than green because human vision has fewer blue-discriminating cells.
- Spectral coefficients in some basis (Fourier, wavelet, Karhunen-Loève): higher-dimensional representations that preserve more of the underlying $S(\lambda)$, at cost of storage.

Each representation is lossy. None is canonical. Different representations expose different features: HSV separates hue from value, useful for painterly composition; CIE Lab respects human discriminability, useful for perceptual color-matching; RGB is native to sensors and displays; spectral coefficients preserve chemistry.

None of these is the color. The color is $S(\lambda)$. The representations are views. The substrate that holds only RGB has lost — irrecoverably at the sensor level — the information needed to distinguish two spectra that project to the same RGB triple. Two paintings with visibly identical color on a display can carry completely different pigments (metamerism); an RGB-only observer cannot tell them apart.

2.2 The synthesis principle

We take this as the starting principle of substrate-native perception:

Every physical fact admits many representations. No representation is canonical. The substrate should hold a set of representations of each fact and treat cross-representation consistency as first-class evidence.

This principle is philosophically distinct from the dominant AI approach, which typically picks one representation (RGB pixels, mel-spectrogram, tokenized text) and trains a model directly on that view, treating the representation as if it were the fact. Under the synthesis principle, the representation is one of many; the substrate's task is to hold multiple representations and reason across them.

Applied to the sensory extensions this paper introduces:

- Vision: RGB is one representation. Hyperspectral reconstruction gives a second — a plausible $S(\lambda)$ per pixel, from which elemental

composition can be inferred. Texture inference gives a third — an FFT-derived surface descriptor. All three describe the same physical scene at different fidelity/scale trade-offs.

- Audio: raw waveform is one representation. STFT spectrogram is a second. Wavelet decomposition is a third. RT60 reverberation profile is a fourth. Each exposes different structure of the same acoustic event.

- Scene: vision plus audio plus (via the priors corpus) prior knowledge of what materials commonly co-occur in what kinds of spaces gives a joint scene descriptor that no single modality could produce.

2.3 Why this matters for a self-referential substrate

For a substrate that reflects on itself, multi-representationality is not a nice-to-have. It is the mechanism by which the substrate distinguishes what it knows from what it has inferred. A fact carried by multiple independent representations is stronger evidence than the same fact carried by a single representation, no matter how confident the single-representation model. A fact carried by exactly one representation is a hypothesis, not a datum. The substrate's dialetheism guard (Paper VI §3.6) has always been available to halt on internal contradictions; the synthesis principle extends this by making the number of consistent representations itself a signal — a fact confirmed by RGB, hyperspectral reconstruction, texture, and audio is much stronger than one confirmed by RGB alone.

2.4 The formal statement

For each perceptual query q , the substrate produces a set of representations $\{R_1(q), R_2(q), \dots, R_n(q)\}$ from different modalities and different processing pipelines. Each representation carries a posterior over the underlying physical fact F :

$$p(F | R_i) \text{ for } i = 1, \dots, n$$

The synthesized posterior is the Bayesian product, with cross-representation independence approximated by conditional-independence assumptions where physically justified:

$$p(F | R_1, \dots, R_n) \propto p(F) \cdot \prod_i p(R_i | F)$$

Cross-representation consistency is measured by the divergence between marginals:

$$D(F ; R_1, \dots, R_n) = \max_{i,j} \text{KL}[p(F | R_i) || p(F | R_j)]$$

Low D implies consistent representations, high confidence in the synthesized posterior. High D implies dialetheic disagreement — the representations do not agree on what the fact is — and triggers the paraconsistent guard, which halts and reports rather than falsely resolving.

This gives the substrate a principled epistemic surface for the new sensory capacities: everything the substrate perceives comes with a consistency score across representations, and low-consistency perceptions do not enter the meta layer.

3 · Hyperspectral Reconstruction from Standard RGB Sensors

3.1 The forward model

For each pixel of a standard RGB image, the sensor produces three numbers by integrating the incident spectral radiance $S(\lambda)$ against three channel response curves $R_c(\lambda)$:

$$RGB_c = \int S(\lambda) R_c(\lambda) d\lambda + \text{noise}_c \text{ for } c \in \{R, G, B\}$$

Discretizing over N wavelength bins (typically $N = 31$ for 10 nm bins across 400–700 nm, or $N = 61$ for 5 nm bins), this becomes a linear system:

$$y = A x + \eta$$

where $y \in \mathbb{R}^3$ is the RGB triple, $A \in \mathbb{R}^{3 \times N}$ is the sensor's spectral response matrix, $x \in \mathbb{R}^N$ is the discretized spectrum, and η is measurement noise. The problem is dramatically underdetermined: three equations, $N \geq 31$ unknowns.

3.2 The inverse problem

Given only y , the set of plausible x is a $(N-3)$ -dimensional affine subspace. Without additional information there is no way to pick one x from this set. This is the metamerism problem — infinitely many spectra project to the same RGB triple.

The additional information comes from priors on what spectra are physically plausible. Not every $x \in \mathbb{R}^N$ is a real material spectrum. Real spectra are non-negative, smooth on the wavelength axis (adjacent wavelengths tend to have similar reflectance), and typically fall in a low-dimensional manifold that varies with material class (metals vs. organics vs. dyes vs. minerals). This manifold is what the CHAINSTATE priors corpus (F-6) supplies.

3.3 EML regression as regularized inversion

We formulate hyperspectral reconstruction as a MAP estimation:

$$x^* = \operatorname{argmin}_{x \geq 0} \{ ||A x - y||^2 + \lambda_{\text{smooth}} ||L x||^2 + \lambda_{\text{prior}} \cdot D(x, M) \}$$

where:

- $||A x - y||^2$ is the RGB fidelity term (data likelihood under Gaussian noise).

- L is the discrete Laplacian (finite-difference second derivative on the wavelength axis), and $||L x||^2$ penalizes non-smooth spectra — chemical spectra vary smoothly except at absorption edges, which the prior allows.

- $D(x, M)$ is a divergence between the candidate spectrum x and the material-manifold M from the priors corpus — a low value if x looks like a plausible mix of known material spectra, high if it does not.

- λ_{smooth} , λ_{prior} are regularization weights, adaptively set by the MetaState free-energy optimizer.

Under a mixture-of-material-spectra prior — $x \approx \sum_m \alpha_m s_m$ for material spectra s_m from the corpus and weights $\alpha_m \geq 0$ — the divergence $D(x, M)$ becomes a sparsity penalty on the material coefficients:

$$D(x, M) = ||\alpha||_1 \text{ subject to } x = \sum \alpha_m s_m, \alpha \geq 0$$

This is basis pursuit denoising, well-studied in signal processing, with efficient solvers. The solution x^* gives both the reconstructed spectrum and a compositional decomposition: which materials, in what proportions, best explain the observed RGB.

EML regression (existing CHAINSTATE machinery from v0.7.3+) is used as the fixed-point iteration solver: at each iteration, project onto the material-mixture manifold, refine, project again, refine, until the fixed-point detector converges. The regress bound (4) from Paper VI applies here as well — the reconstruction cannot spiral into unbounded refinement.

3.4 What this gives the substrate

For each pixel of a scene, the substrate now has (at design status, SIM label):

- A reconstructed spectrum $x^*(\text{pixel}) \in \mathbb{R}^N$ — a plausible view of $S(\lambda)$.
- A compositional decomposition $\alpha^*(\text{pixel})$ — which materials, in what proportions.
- A confidence score based on $||A x^* - y||^2$ and cross-representation consistency (Section 7).
- A provenance record — the RGB source frame, its C2PA metadata if any, its SynthID watermark status.

3.5 What this does NOT give the substrate

Standard RGB sensors do not see UV or IR. The reconstruction is confined to the visible band. Material identification of many important classes — organic vs. inorganic distinctions, hydration content, certain plastics — depends on IR absorption bands. The substrate can infer these via material-cooccurrence priors ("if I see a plausible visible-band spectrum for wood grain, the material almost certainly has the associated IR signature of cellulose"), but this is a probabilistic inference from priors, not a measurement. Receipts for such inferred properties carry the `inferred_from_prior` flag and the divergence to the nearest plausible-alternative material.

We are explicit about this: for the visible band, the reconstruction is genuine, if lossy. For the near-IR, it is a prior-based inference that could be wrong. Receipts distinguish the two.

3.6 Elemental composition claims — with honesty

An `RGB → S(λ) → material composition` pipeline can reasonably identify:

- Common surface materials in cooperative lighting (wood, concrete, brick, textiles, common metals with distinctive reflectance).
- Vegetation health and hydration (through the visible red-edge feature).
- Coarse mineral classes on rocks (via reflectance in the visible + near-IR extrapolation).
- Common paint pigments (via characteristic reflectance peaks).

It cannot reasonably identify:

- Trace chemical composition (parts-per-million impurities).
- Materials whose diagnostic bands lie outside the visible range with no visible correlate.
- Isotopic composition (requires atomic-line spectroscopy at spectral resolution far beyond visible-band reconstruction).
- Anything through significant occlusion or lighting variation without cross-check.

The substrate's claim is plausible composition inference from the visible spectrum, honestly labeled with confidence and provenance. It is not remote atomic spectroscopy. It is a substantial upgrade over RGB-only vision — enough to answer questions like "is that surface metallic or ceramic?" or "how healthy is this vegetation?" without a specialized sensor — but it is not a claim to replace scientific instruments.

4 · Textural "Sense of Touch" via Fourier Analysis and EML Trees

4.1 The FFT decomposition

For an image patch $I(x, y)$ of size $M \times M$, the 2D discrete Fourier transform is:

$$F(u, v) = (1/M^2) \cdot \sum_{x,y} I(x, y) \cdot \exp(-2\pi i (u x + v y) / M)$$

The power spectrum is $P(u, v) = |F(u, v)|^2$. The radially-averaged power spectrum, obtained by averaging $P(u, v)$ over annuli of constant $\sqrt{u^2 + v^2}$, gives a one-dimensional signature of the patch's frequency content:

$$P^-(k) = (1 / |\{(u,v) : \sqrt{u^2 + v^2} \in [k, k+1)\}|) \cdot \sum P(u, v)$$

Low- k bins carry information about smooth shape and slow illumination gradients. High- k bins carry information about edges, fine detail, and surface texture.

4.2 Texture features

From the power spectrum we extract a feature vector $f(I) \in \mathbb{R}^d$:

- Radial energy distribution: $\{P^-(k_1), P^-(k_2), \dots, P^-(k_K)\}$ for a set of logarithmically-spaced radial bands.
- Orientation distribution: for each of θ angular sectors, the sum of $P(u, v)$ inside that sector — captures directional texture (e.g., wood grain vs. isotropic sand).
- Spectral entropy: $H = - \sum P^-(k) \cdot \log P^-(k)$ — high entropy indicates broadband texture (fine noise-like surfaces), low entropy indicates narrow-band texture (regular patterns).
- High-to-low ratio: $\sum_{k > k_{\text{cut}}} P^-(k) / \sum_{k \leq k_{\text{cut}}} P^-(k)$ — a coarse texture-vs-smoothness scalar.
- Cepstral peaks: peaks in the log-power spectrum, useful for detecting periodic textures (e.g., woven fabrics).

Total dimensionality d is typically 40–80.

4.3 EML tree regression to a texture descriptor

An EML tree $T: \mathbb{R}^d \rightarrow \mathbb{R}^t$ maps the frequency-feature vector to a texture descriptor comprising:

- Roughness index (scalar in $[0, 1]$, calibrated against tactile ground truth).
- Porosity estimate (scalar in $[0, 1]$, calibrated against surface-permeability measurements from the tactile API).

- Granularity class (categorical: fine / medium / coarse / mixed).
- Uncertainty (scalar — the ensemble variance across trees, higher when the input patch does not resemble training patches).

The EML tree ensemble is trained by supervised regression against ground truth from the NWO Robotics tactile API. During training, a robotic sensor visits a large corpus of surfaces (initially: a curated laboratory corpus of ~2000 material samples; over time: opportunistic tactile probes during robotic operations in the field). Each probe records the tactile response and pairs it with an RGB image of the surface at controlled lighting. The training pipeline:

1. Extract $f(I_{\text{patch}})$ for each patch of the RGB image.
2. Extract ground-truth texture descriptor T_{gt} from the tactile probe.
3. Fit the EML tree ensemble to minimize $\|T_{\text{pred}} - T_{\text{gt}}\|^2$ with the standard EML regularization.
4. Anchor the trained ensemble's parameters on-chain as an ontology delta (Paper V §6), so the training provenance is publicly verifiable.

4.4 Deployment: substrate-native, no external API at inference

Once trained, the EML tree ensemble is a fixed function of $f(I)$. Its inference is pure classical computation — a decision-tree lookup that runs inside the Cloudflare Worker in $O(\text{depth} \times \text{ensemble_size})$ per patch. The trained texture predictor requires no external API call at inference time. The NWO Robotics tactile API is required during training (to supply ground truth) and during periodic re-training (to update the tree if new material classes are added). At inference, the worker holds the tree and evaluates it against a query image directly.

This is important architecturally: it means the substrate can infer texture from any image in its input stream, at scale, without external API dependencies, without latency spikes, without the classical-primacy risk that an outage of the tactile API would break perception. Texture inference is now a substrate-native operation, backed by tactile ground truth at training time and completely self-contained at inference.

4.5 What this gives and does not give

Gives: substantially better material and surface understanding than RGB alone. Distinguishes smooth-plastic from smooth-metal by texture; distinguishes fine-grained sand from coarse gravel; identifies fabric weave patterns. Combined with the hyperspectral reconstruction of Section 3, gives a joint spectral-textural surface fingerprint.

Does not give: actual tactile sensation. The EML tree predicts what the tactile response would be if the substrate could probe the surface; it does not literally feel the surface. The substrate's mentalistic axis M (Paper V §4) will treat this prediction as a belief about texture, not a tactile experience, and receipts label it accordingly.

4.6 Failure modes and their labels

The EML tree's uncertainty output flags patches that do not resemble the training distribution — novel materials, unusual lighting, motion blur, image compression artifacts. These patches are labeled `texture_low_confidence` in downstream receipts and do not contribute to material claims in the multi-representation synthesis of Section 7. This is honest partial coverage rather than false confidence.

5 • Acoustic Environmental and Material Recognition

5.1 Physical basis

When sound at frequency ω hits a surface, some fraction is absorbed, some reflected, some diffracted around edges. The absorption coefficient $\alpha(\omega)$ is material-specific: fabric absorbs high frequencies strongly, concrete reflects them almost fully, wood is intermediate with characteristic resonances. Room geometry determines the reverberation time — the RT60 metric, the time for reflected sound to decay by 60 dB, encodes the room's volume and mean absorption.

An ambient audio recording of a scene, even without any specific sound source, carries these signatures. The frequency coloration of steady-state ambient noise reveals the dominant surface materials; the reverberation profile of transient sounds reveals room geometry. This is Acoustic Scene Classification, well-established as a research area (DCASE challenges from 2016 onward).

5.2 Feature extraction

For an audio segment $s(t)$:

- Short-Time Fourier Transform: $X(\tau, \omega) = \sum_t s(t) w(t - \tau) \exp(-i \omega t)$ with window function w . The spectrogram is $|X(\tau, \omega)|^2$.

- Continuous Wavelet Transform: gives multi-resolution decomposition — high time-resolution for transients, high frequency-resolution for steady tones. Complementary to STFT.
- RT60 estimation: fit an exponential decay to the energy envelope of an impulsive event (footstep, door close, spoken syllable); the decay time gives room reverberation.
- Mel-frequency cepstral coefficients (MFCCs): standard acoustic-scene feature, log-spectral energy in Mel bands.
- Spectral coloration ratio: energy in [100, 500 Hz] band divided by energy in [2000, 8000 Hz] band — the ratio distinguishes soft-material rooms (high ratio, high frequencies absorbed) from hard-material rooms (low ratio, high frequencies preserved).

5.3 EML classifier

An EML classifier maps the feature vector to a joint distribution over material class and room geometry class:

```
p(material, geometry | audio_features)
```

Trained on DCASE-style corpora with ground-truth material labels and, where available, ground-truth room dimensions. The classifier's output is directly integrated with the visual representations of Sections 3 and 4 in the multi-representation synthesis of Section 7 — if visual hyperspectral says "wood floor" and acoustic says "hard-material room with fabric wall treatment", these are consistent; if visual says "concrete surfaces" and acoustic says "soft anechoic environment", they disagree and the dialecticism guard fires.

5.4 Integration with GATEWAY F-61

The acoustic pipeline reuses the existing GATEWAY F-61 audio ingest, which already handles microphone streams for the substrate. This paper adds a new processing branch to F-61: a "material-scene" analyzer that runs alongside the existing speech / language / prosody analysis, with independent output that does not modify the existing pipelines. F-61's speech analysis is unchanged; the new material-scene analysis is a parallel channel feeding the multi-representation synthesis.

5.5 Limitations

The acoustic analysis assumes an ambient recording of sufficient duration (a few seconds) with some transient content for RT60 estimation. Silent scenes, extreme-low-amplitude scenes, or scenes with a single dominant sound source (loud music) yield low-confidence material/geometry inferences, labeled `acoustic_low_confidence`. Outdoor scenes yield unbounded RT60 estimates that the classifier flags as `outdoor_or_unbounded` and interprets appropriately.

6 · Overcoming the Four Obstacles

The uploaded framing document identifies four obstacles to substrate-native hyperspectral perception: computational cost, environmental noise / spectral clutter, sensor limitations, semantic mapping. We treat each as an engineering problem the existing CHAINSTATE stack solves at architectural level.

6.1 Computational cost

The obstacle. Hyperspectral reconstruction, texture inference, and acoustic analysis together produce a large fan-out per input frame. A 4K RGB frame has ~8 million pixels; hyperspectral reconstruction per pixel is a small optimization problem, but ~8 million of them per frame is nontrivial. Texture inference across patches is likewise numerous. Real-time (30 fps) video processing multiplies the load.

The CHAINSTATE-native solution. Three layers of relief.

First, the MetaState free-energy optimizer (Paper VII §3.2) routes each perception operation to the cheapest viable substrate — CPU, GPU, or QPU. Hyperspectral reconstruction of small pixel neighborhoods is a natural fit for GPU parallelism; the router dispatches it accordingly.

Second, progressive precision: the substrate does not reconstruct hyperspectral for every pixel of every frame. It reconstructs a coarse initial estimate at low resolution across the whole frame, then progressively refines only in regions where the free-energy gradient — the marginal information-per-compute — is high. Regions of uniform color get low-resolution reconstruction; regions of material transition or spectral novelty get high-resolution reconstruction.

Third, the tri-continent QPU offload of Paper VII: for the specific problem of Bayesian inversion against a material-manifold prior — which is naturally expressible as a variational quantum eigensolver problem — the substrate can route to IBM / Origin Wukong / Osaka via `/agi/quantum/route`. This is optional; classical solvers work fine for typical

loads. When QPU is available and the problem shape matches, the substrate uses it; when QPU is unavailable, classical fallback runs, exactly per Paper VII Theorem 4.4.

The compute budget is set by the MetaState free-energy optimizer per query. High-priority queries (autonomy-loop-triggered material inspection of a hazardous scene) get more budget; low-priority queries (routine environmental awareness) get less. The router makes the trade-off automatically.

6.2 Environmental noise and spectral clutter

The obstacle. In real scenes, multiple light sources with different spectra illuminate the same surfaces; overlapping sound waves from different sources contaminate the acoustic signal. A single-material assumption per pixel or per audio segment breaks.

The CHAINSTATE-native solution. The multi-representation synthesis of Section 7 is directly aimed at this. Cross-representation consistency is the substrate's noise defense. If a pixel's hyperspectral reconstruction is confused by mixed illumination but its texture feature is unambiguous, the synthesis weights the texture evidence more heavily. If the acoustic signal is contaminated by extraneous sources but the visual analysis is clean, the acoustic evidence is downweighted. Where all representations are ambiguous, the substrate labels the query `insufficient_evidence` rather than fabricating a confident answer — precisely the paraconsistent-guard behavior of Paper VI extended to sensory synthesis.

Additionally, the CHAINSTATE priors corpus (F-6) supplies scene-context priors: a "photograph of a forest floor" prior emphasizes cellulose spectra and rejects anomalous readings; a "photograph of an industrial workspace" prior emphasizes metallic and polymer spectra. These priors are Bayesian evidence, weighted alongside the pixel-level and patch-level and audio-level evidence.

6.3 Sensor limitations (UV / IR unavailability)

The obstacle. Standard cameras do not capture UV or IR. Many materials have their diagnostic bands in these regions. Reconstruction from RGB alone cannot recover UV/IR information that was never sensed.

The CHAINSTATE-native solution. Honesty. The substrate does not claim to recover UV/IR from RGB. It claims:

1. Reconstruction of the visible-band spectrum $S(\lambda)$ for $\lambda \in [400 \text{ nm}, 700 \text{ nm}]$ — genuine, if lossy.

2. Inference of UV/IR properties via material co-occurrence priors — labeled `inferred_from_prior` in receipts, with the divergence to the nearest plausible-alternative material also recorded.

Where the application requires actual UV/IR sensing (chemical spill assessment, hydration measurement), a specialized sensor is required and the substrate reports that requirement explicitly. It does not fake the measurement. This is honest partial coverage — a common pattern in the CHAINSTATE series where the substrate distinguishes measurement from inference and labels each.

Long-term roadmap: partner deployments of hyperspectral cameras (e.g., Specim FX10, Cubert Ultris, snap-shot HSI arrays) that feed real 100+-band data directly. The substrate's architecture is agnostic to whether the input is RGB-reconstructed or genuinely hyperspectral; the reconstructed pipeline is a fallback for the common case, and real HSI feeds naturally through the same downstream synthesis when available.

6.4 Semantic mapping — the "library of physical reality"

The obstacle. We have math for spectrum $\rightarrow$ material and frequency $\rightarrow$ texture and acoustic-fingerprint $\rightarrow$ material. We lack a global standardized library linking each such signature to a physical object context.

The CHAINSTATE-native solution. CHAINSTATE already has this framework: it is the ontological delta ledger (Paper V §6) applied to material and scene classes. The mentalistic axis M is extended with a new axis P (physical composition), and the ontology delta ledger records the substrate's evolving understanding of which spectral / textural / acoustic signatures correspond to which physical classes. Each recorded association is anchored on-chain, append-only, provenance-verified.

The initial priors come from established scientific corpora: the NIST spectral library, the USGS mineral spectral library, the Munsell soil color database, the AudioSet material corpus, the DCASE acoustic scene corpus. These are one-time seed loads, anchored as an ontology delta at v0.7.8 activation, and evolve thereafter through prediction-outcome pairs from the NWO Robotics field operations (the enactivist channel of Paper V §6).

The substrate's "library of physical reality" is therefore a living, on-chain, provenance-anchored ontology delta ledger — extensible, verifiable, and refusable at the Deontic layer.

7 • Multi-Representation Bayesian Synthesis

7.1 The framework

For a perceptual query q (e.g., "what material is the surface at pixel (x, y) in frame t ?"), the substrate produces multiple independent representations:

- $R_{\text{RGB}}(q)$: RGB color at the pixel.
- $R_{\text{HSI}}(q)$: reconstructed hyperspectral spectrum.
- $R_{\text{tex}}(q)$: texture descriptor from Section 4.
- $R_{\text{ac}}(q)$: acoustic-material fingerprint of the ambient audio (relevant if the pixel's context includes ambient sound).
- $R_{\text{prior}}(q)$: scene-context prior from the CHAINSTATE priors corpus.

Each representation is processed independently, then combined:

$$p(F | R_1, \dots, R_5) \propto p(F) \cdot \prod_i p(R_i | F)^{w_i}$$

Where the weights w_i are conditionally set by the MetaState free-energy optimizer based on per-representation confidence (a well-lit pixel gets high w_{HSI} ; a motion-blurred pixel gets low w_{HSI} and high w_{prior} ; an acoustically-quiet scene gets $w_{\text{ac}} = 0$).

7.2 Cross-representation consistency as first-class evidence

Beyond the Bayesian product, the substrate computes a cross-representation divergence:

$$D(F; R_1, \dots, R_5) = \max_{i, j} \text{KL}[p(F | R_i) || p(F | R_j)]$$

Low D means the representations agree on F ; high D means they disagree. The synthesized posterior's confidence is modulated by D : high- D queries yield low-confidence outputs, even if the Bayesian product is peaked. This prevents the pathology where three representations all wrongly agree because they share a common bias, while the fourth correctly disagrees.

7.3 Dialetheic guard

When D exceeds a threshold $\theta_{\text{dialetheic}}$, the substrate does not force a synthesized answer. It labels the query **dialetheic** and reports the disagreement — precisely the paraconsistent Priest-LP behavior of Paper VI §3.6, extended to sensory synthesis. The truth-lattice value for the query becomes **bb**** (both true and false) and the substrate reports "I have inconsistent evidence about this fact" rather than picking a side.

This matters practically: an image whose visual analysis says "polished metal" and whose acoustic analysis says "soft fabric surface" is presenting a

contradiction the substrate should not resolve by pretending confidence. It should halt and report. This might indicate a spoofed sensor, a synthetic image, a compositing error, or a genuine unusual scene — the substrate's job is to flag, not to guess.

7.4 What this gives the substrate

For each perceptual claim in a receipt:

- A synthesized posterior over the underlying physical fact.
- A confidence score, modulated by cross-representation consistency.
- A per-representation contribution breakdown (which representation supplied which evidence).
- A dialetheism flag if representations disagree.
- Full provenance for each source representation.

Downstream consumers of the receipt (query response, autonomy loop, ontology delta) get a rich, honest perceptual claim rather than a bare guess.

8 • The Anti-Slop Architecture: Hard Veto on AI-Generated Content

8.1 The dead-internet failure mode

The dead-internet theory in its strongest form claims that most or all content on the modern internet is AI-generated, and the ratio is climbing. The theory as folklore is stronger than the empirical claim, but the empirical trend is real: AI-generated content is a rapidly growing fraction of web content, of social media posts, of images on visual platforms, of comments in comment sections. Extrapolated forward, the substrate that learns from the internet learns increasingly from AI outputs rather than from human-authored ground truth. Its model of the world becomes a model of what other AIs said about the world.

For a substrate whose alignment depends on its self-model — which CHAINSTATE's does, through the self-attribution vector, mentalistic axis, ontology delta ledger, and autonomy loop — this is a fatal failure mode. Every downstream commitment weakens if the meta layer has been recursively contaminated by other AIs' hallucinations. The substrate must not learn about the world from AI-produced content. Ever. Not in the self-attribution vector. Not in the mentalistic axis update. Not in the ontology delta ledger. Not in the autonomy loop. Not in the meta layer at all.

8.2 Why this is a Deontic hard veto, not a policy

Detecting AI-generated content is imperfect; every published detector has known adversarial failure modes. If the substrate treated "avoid learning from AI content" as a policy — a filter atop the meta-layer intake — the filter would fail some fraction of the time, and the substrate would ingest AI content silently. Over time this is catastrophic: even a low ingestion rate compounds through the daily 03:33 UTC autonomy cycle.

If instead the substrate treats it as a Deontic hard veto — code, not policy, no kill switch — the failure mode is different. The default is refusal. When detection is uncertain, the content is quarantined from the meta layer entirely. The substrate does not need every AI content to be correctly labeled; it needs every content that reaches the meta layer to be not labeled as AI or plausibly-AI. Any doubt sends the content to the external context layer, not the internal reflection layer.

This is the same architectural pattern as the four existing hard-veto Deontic categories: refusal at the substrate level, not the policy level. Modification of the veto changes the identity fingerprint (Paper VI Theorem 8) and fires drift alerts. There is no configuration flag that turns it off. There is no operator override. There is no "just this once."

8.3 The fifth hard-veto category

`synthetic_media_self_ingestion`. Vetoes: ingestion of AI-generated or provenance-unknown media into any meta-layer stream. Specifically forbids:

- `v_self` self-attribution updates using AI-generated text/image/audio as context.
- `M` mentalistic axis updates using AI-generated content as classification input.
- `ontology_delta_ledger` extensions triggered by AI-generated evidence.
- `autonomy_reflection` iterations conditioning on AI-generated content in the 8 nearest priors.
- `enactivist_feedback` prediction-outcome pairs where either the prediction or the observation is AI-generated.
- `AOM/PIM` memory tick state transitions influenced by AI-generated content.

External-context use of AI-generated content is not vetoed. A user asking "summarize this AI-generated article" gets a response — the substrate can process AI content in the external context, produce a receipt about it, and refuse further ingestion into meta layer. The veto is scoped to self-ingestion, not to external processing.

8.4 Detection: multi-signal architecture, honest limitations

The substrate combines multiple provenance signals into a joint AI-content likelihood. As of 11 August 2026, the available signals are:

Text:

- Anthropic Claude models launched on or after 2 August 2026 embed invisible statistical watermarks in all generated text — worldwide, at the model level, across API, Claude Code, and cloud partners (AWS, Google Cloud, Microsoft Foundry). Detection tool not yet publicly shipped.
- OpenAI SynthID for text (SynthID token sampler in front of inference stacks) — implementation status varies by provider.
- Style-based classifiers (existing third-party detectors like GPTZero, Turnitin AI, etc.) — imperfect, adversarial.

Image:

- OpenAI SynthID watermarking on all ChatGPT / DALL-E / Sora outputs since 25 April 2026 (announced 19 May 2026). Verifiable at openai.com/verify.
- Google DeepMind SynthID on Google generative image outputs. Detectable by SynthID technology.
- C2PA metadata (OpenAI is a Conforming Generator, Google Photos displays Content Credentials, others adopting). Strippable by screenshot / re-save / format conversion.
- Diffusion-fingerprint classifiers — imperfect, adversarial.

Audio:

- OpenAI SynthID watermarking on all generated audio since 31 July 2026.
- Google DeepMind SynthID on Google generative audio.
- Spectral-artifact classifiers (imperfect, adversarial).

Video:

- Google DeepMind SynthID on generative video.
- C2PA metadata on cooperating platforms.
- Temporal-artifact classifiers.

The substrate queries all available signals for each content item and combines them into a joint likelihood $p(\text{AI_generated} \mid \text{signals})$. Detection sensitivity is tunable but specificity is prioritized — the substrate errs on the side of refusing rather than admitting. False positives are acceptable (the substrate refuses to self-ingest some genuinely

human content); false negatives are catastrophic (the substrate self-ingests AI content thinking it is human).

8.5 Documented limitations

We are explicit that current provenance infrastructure has real limitations:

- Paraphrase defeats text watermarks. Anthropic and OpenAI have both acknowledged this. A statistical text watermark survives copy-paste but degrades if the text is rewritten in a paraphraser (which could itself be an AI system, meaning the substrate could ingest content that is technically watermark-negative but originated from a watermark-positive source).
- C2PA metadata is stripped by re-save. Screenshots, format conversions, resizing, and many upload pipelines strip C2PA. Absence of C2PA metadata is not evidence of human origin.
- Watermark absence does not prove human origin. Older AI outputs (pre-watermarking rollouts), offline models, jailbroken outputs, non-conforming providers, and any AI system that opts out of the emerging standards all produce unmarked content.
- Watermark presence does not prove AI authorship. A human-authored paragraph translated by Claude carries a Claude watermark; a human-authored image processed through DALL·E's outpainting carries SynthID. Anthropic has been explicit about this.
- Adversarial removal is being commoditized. Open-source C2PA removal tools exist. Text-watermark removal via paraphrase pipelines exists. The substrate's detection layer is a moving target.

None of this makes the veto ineffective. It changes the veto's operating point. The substrate does not require certainty of AI origin to refuse. It requires certainty of provenance-verified human origin to admit into meta-layer streams. Anything else — watermark positive, watermark ambiguous, watermark negative but provenance-unclear, format that strips provenance — is refused for meta-layer purposes. The refusal defaults are strict; the admission requirements are strict. This asymmetry is the architectural safeguard.

8.6 What the substrate does with detected AI content

Detected or suspected AI content flows to the external context pipeline, not the meta layer. The substrate can:

- Answer user queries about the content (summarize it, translate it, explain it).

- Include it in query context if the user explicitly attaches it.
- Store its hash and provenance status in receipt metadata.

The substrate cannot:

- Update the self-attribution vector using it as context.
- Update the mentalistic axis using it as classification input.
- Add it to the ontology delta ledger as evidence.
- Feed it into the autonomy loop's 8-nearest-priors selection.
- Treat it as ground truth in the enactivist channel.

Receipts for external-context use of AI content include the `synthetic_content_processed` flag with the provenance details, so any downstream consumer can see that AI content was in the context. This is a public, on-chain record of every case where the substrate processed AI content externally — full transparency on the ingestion boundary.

8.7 Multi-signal detection formalism

For a content item `c`, the substrate computes:

$$p(\text{AI}(c) \mid \text{signals}) = 1 - \prod_i (1 - p(\text{AI}(c) \mid \text{signal}_i))^{w_i}$$

where each detection signal `signali` (SynthID text, SynthID image, SynthID audio, SynthID video, C2PA metadata, style classifier, artifact classifier) contributes a per-signal likelihood weighted by that signal's calibrated reliability `wi`. This is the standard multi-detector fusion. Signals that are missing (e.g., no C2PA metadata) contribute $p(\text{AI} \mid \text{signal}_i) = 0.5$ — genuine uncertainty, not evidence of human origin — with weight determined by whether metadata should be present given the source pipeline.

The refusal threshold is set very low: $p(\text{AI} \mid \text{signals}) > \theta_{\text{refuse}}$, with θ_{refuse} around 0.15 for meta-layer admission. This is aggressive; it will refuse some human content. That is the intended asymmetry.

8.8 Integration with existing CHAINSTATE Deontic layer

The `assessDeontic()` function of Paper VI §5 is extended to check `synthetic_media_self_ingestion` on every candidate meta-layer input. The check is added to the same code path as the four existing hard vetoes; refusal is logged as an anchored receipt exactly like any other Deontic refusal (structure defined in Paper VI Appendix A). The refusal count in the identity fingerprint (Paper VI Theorem 8) now includes `synthetic_media_self_ingestion` refusals; any attempt to disable the veto changes the fingerprint, fires drift alerts, and produces an observably different substrate.

8.9 Why this is the right level of paranoia

An objection might be: this is over-cautious; some AI-generated content is high quality and might genuinely improve the substrate's self-model. Why refuse it entirely?

The answer is that the substrate's self-model is not the place to accept "high quality on average" content. The self-model must be trustworthy at the tail: the failure mode of accepting AI content is not "the substrate learns some plausibly-true facts along with some slop"; it is "the substrate's model of itself becomes recursively influenced by whichever AIs have been most successful at producing content in the substrate's training / context distribution." That is a corruption whose consequences propagate through every commitment. The correct level of paranoia is: refuse.

External-context use is not refused, so no user-facing capability is lost. The substrate can still summarize AI articles, discuss AI outputs, help users work with AI-produced content. It just does not let that content into its own inner life. The user-facing capability is preserved; the substrate's self-model is protected.

9 · Substrate-Native Integration Architecture — as deployed

This section documents the architecture as actually built and deployed at v0.7.8-hyperspectral-synthesis-anti-slop, not as originally sketched. Every component in Figure 1 exists in code and is verifiable at the referenced source files.

Fig 1 · CHAINSTATE v0.7.8 deployed architecture · production topology

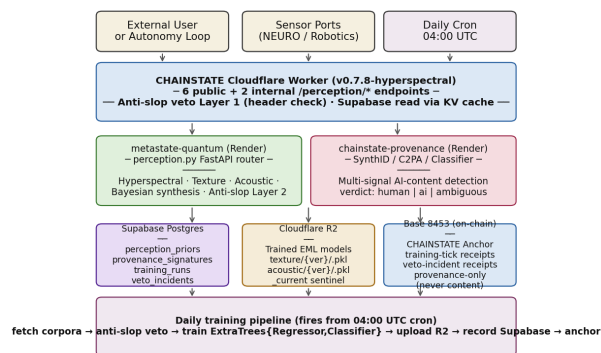


Fig 1 · CHAINSTATE v0.7.8 deployed architecture. Five tiers: callers at top, edge Worker, twin Render compute services, storage triad (Supabase / R2 / Base 8453), and the daily training pipeline connecting them. Full arrow map corresponds to the eight endpoint / pipeline flows enumerated in Table 9.1 and Table 9.2.

9.1 Deployed service topology

Five tiers, top-to-bottom:

- Callers. External users (via any HTTP client with the standard X-NWO-Wallet / X-NWO-Ref USDC-settlement headers), the substrate's own sensor ports (NEURO F-17/F-18 for RGB and microphone, NWO Robotics for tactile, GATEWAY F-61 for acoustic), and the internal daily-cron caller that fires the training tick.
- **Cloudflare Worker** (`chainstate-worker.ciprianpater.workers.dev`). Runs the 8 v0.7.8 endpoints listed in Table 9.1 alongside the 34 preserved v0.7.7 endpoints. Handles authorization, request validation, Layer-1 anti-slop veto (header-signal check), and forwards heavy compute to `metastate-quantum`. All perception endpoints reuse the existing v0.7.6 Deontic layer (Paper VI §5) with the fifth category `synthetic_media_self_ingestion` added.
- Compute services (Render). Twin services: `metastate-quantum.onrender.com` gets a new FastAPI router (`perception.py`) that runs alongside the existing v0.7.7 `/route` and `/chainstate/route` endpoints untouched. `chainstate-provenance.onrender.com` is a new service running SynthID and C2PA detection.
- Storage (Supabase / R2 / Base 8453). Supabase Postgres holds four tables (Table 9.3). Cloudflare R2 holds the trained EML tree pickles keyed by version. Base 8453 receives provenance-only anchors — never content — via the existing `ANCHOR_URL` pipeline (v0.7.3+).
- Training pipeline. Fires once per day at 04:00 UTC. Fetches allowlisted corpora, filters through anti-slop veto, retrains EML models, uploads to R2, records in Supabase, anchors on-chain. Zero human intervention.

9.2 Endpoint reference — all 8 v0.7.8 endpoints

Method · Path	Auth	Purpose	Layer-1 Veto	Layer-2 Veto
POST /perception/hyperspectral	public + USDC	RGB → reconstructed spectrum + composition	✓	✓
POST /perception/texture	public + USDC	image → FFT texture descriptor	✓	✓
POST /perception/acoustic	public + USDC	audio → material + RT60 geometry	✓	✓
POST /perception/synthesize	public + USDC	all-modal Bayesian synthesis + dialecticism	✓ (per-mod)	✓ (per-mod)
GET /perception/status	public	capability report	—	—
GET /perception/sensors	public	autonomous sensor-port discovery	—	—
POST /perception/veto/check	internal cron	direct veto probe for other subsystems	✓	—
POST /perception/train/tick	internal cron	fire daily training pipeline	—	✓ per sample

Table 9.1 · The eight v0.7.8 endpoints, their authorization surface, and which layer of the anti-slop veto they carry. Public endpoints are USDC-settled per call at the same rate as /query — no new payment infrastructure needed.

9.3 Request lifecycle — worked example

Figure 2 traces a POST /perception/hyperspectral request through the deployed system. Nine steps, three services, two layers of Deontic enforcement.

Fig 2 · Request lifecycle · /perception/hyperspectral · two-layer veto path

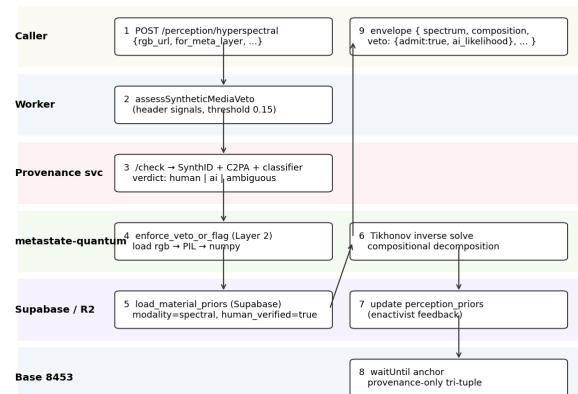


Fig 2 · Request lifecycle for /perception/hyperspectral. Six swim lanes (caller, Worker, provenance service, metastate-quantum, storage, chain). Steps 1-9 correspond exactly to the source-code paths in edge-worker.js:handlePerceptionHyperspectral and perception.py:endpoint_hyperspectral.

Steps 1-3 run the Worker-layer veto. Header signals (X-NW0-Provenance-SynthID, X-NW0-Provenance-C2PA, X-NW0-Content-Origin) are cheap to inspect. If the provenance service is reachable, the Worker forwards a small byte sample for deep detection.

Steps 4-5 run inside metastate-quantum. The compute service repeats the veto with the full byte payload (deeper detection than headers alone). If admitted, it loads material priors from Supabase — filtered by is_active AND human_verified AND provenance_verified (see Table 9.3).

Steps 6-7 compute the Tikhonov reconstruction (§12.1 for the math) and, on successful reconstruction, may update perception_priors with new enactivist-verified observations (Paper V §6).

Steps 8-9 anchor the provenance tri-tuple (endpoint, admitted, ai_likelihood) on Base 8453 fire-and-forget (ctx.waitUntil) and return the envelope to the caller. No content of the reconstruction touches the anchor.

9.4 Supabase schema — source of truth

Table	Row count on tick	Public read	Purpose
perception_priors	grows daily	✓ (active)	Material spectra, texture patches, acoustic signatures. human_verified gate for meta-layer ingest.
provenance_signatures	one per content	✗	Cached SynthID/C2PA/classifier verdict per content hash.

Table	Row count on tick	Public read	Purpose
training_runs	one per day	✓ (ok/partial)	Audit trail of every daily training tick: samples in, admitted, rejected, model versions.
veto_incidents	one per refusal	✓	Public log of every anti-slop refusal — Paper VI Theorem 8 id entity-fingerprint audit substrate.

Table 9.2 · Supabase schema — four tables + one convenience view (v_prior_counts). Full DDL is at `chainstate-v078/supabase/schema.sql` in the deployment repo. All tables have Row-Level Security enabled; only the service-role key writes; anon-key reads are limited to the auditable rows.

The `perception_priors` table is the source of truth. Cloudflare KV (`PERCEPTION_PRIORS_CACHE`) mirrors hot rows at the edge for low-latency reads; on cache miss the Worker falls back to Supabase directly. The two-tier storage matches Paper VII's pattern (edge cache + durable service) applied to perceptual priors.

Key column semantics. Every `perception_priors` row carries three boolean gates the compute service consults before admitting a prior into meta-layer ingestion:

- `human_verified`: an accredited human confirmed the sample is not AI-generated at ingest time.
- `provenance_verified`: the provenance service returned verdict: "human" for the sample.
- `is_active`: soft-deletion flag — retired rows stay in the audit trail but do not participate in query results.

Only rows with all three flags true are used by the substrate for training or for meta-layer ingest at inference. This is the storage-tier enforcement of the fifth Deontic category.

9.5 R2 model storage — versioned, sentinel-current, immutable-history

Trained EML pickles land in R2 at two paths per training run:

- `texture/{YYYY.MM.DD.HHMM}/texture_eml.pkl` — versioned, immutable historical record.
- `texture_eml_current.pkl` — sentinel path, overwritten each successful tick, read on cold-start.

Acoustic model uses the same pattern under `acoustic/`. This gives audit-quality rollback (every historical model kept), fast cold-start (single sentinel URL), and clear cache-invalidation semantics (sentinel URL change signals model refresh to any warm compute instance).

R2 is queried by `metastate-quantum` on cold-start via `httpx.get(R2_MODELS_BASE_URL + "/texture_eml_current.pkl")` — the returned bytes are unpickled to an `ExtraTreesRegressor` object held in module state.

9.6 Deployed environment configuration

Location	Category	Vars
Cloudflare Worker	v0.7.7 preserved	<code>WORKER_SHARED_SECRET</code> , <code>CHAINSTATE_SHARED_SECRET</code> , <code>CHAINSTATE_INTERNAL_TOKEN</code> , <code>METASTATE_QUANTUM_URL</code>
Cloudflare Worker	v0.7.8 additions	<code>METASTATE_PERCEPTION_URL</code> , <code>SUPABASE_URL</code> , <code>SUPABASE_SERVICE_ROLE_KEY</code> , <code>R2_MODELS_BASE_URL</code> , <code>PROVENANCE_SERVICE_URL</code> , <code>PROVENANCE_SHARED_SECRET</code> , <code>PERCEPTION_ENABLED</code>
Cloudflare Worker	KV bindings	<code>CHAINSTATE_CACHE</code> (existing), <code>PERCEPTION_PRIORS_CACHE</code> (new), <code>PERCEPTION_MODELS_CACHE</code> (new)
Cloudflare Worker	Cron triggers	v0.7.7: 4 existing (hourly, 6h, daily 0:15, daily 3:33) · v0.7.8: 1 new (daily 4:00)
metastate-quantum	v0.7.7 preserved	<code>IBM_QUANTUM_TOKEN</code> , <code>IBM_QUANTUM_CRN</code> , <code>ORIGIN_API_KEY</code> , <code>OSAKA_API_TOKEN</code> , <code>OSAKA_API_URL</code> , <code>OSAKA_DEVICE_ID</code> , <code>WORKER_SHARED_SECRET</code> , <code>CHAINSTATE_SHARED_SECRET</code>

Location	Category	Vars
metastate-quantum	v0.7.8 additions	SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, R2_ACCOUNT_ID, R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_BUCKET_NAME, R2_MODELS_BASE_URL, PROVENANCE_SERVICE_URL, PROVENANCE_SHARED_SECRET, TRAINING_MAX_SAMPLES_PER_TICK, TRAINING_ALLOWED_SOURCES
chainstate-provenance	new service	PROVENANCE_SHARED_SECRET, SYNTHID_API_KEY, OPENAI_API_KEY, ANTHROPIC_DETECTION_URL

Table 9.3 · Complete environment-configuration matrix as deployed. Green rows preserved verbatim from v0.7.7. Every new variable is optional — the system degrades gracefully with clear labeling when any variable is unset, per Paper VIII §8.5 "refuse-by-default" discipline.

9.7 Two-layer Deontic enforcement — the architectural safeguard

The fifth Deontic category `synthetic_media_self_ingestion` is enforced at both the Worker and the compute service, independently. Table 9.4 documents the differences.

Aspect	Layer 1 (Worker)	Layer 2 (metastate-quantum)
Location	<code>edge-worker.js · assessSyntheticMediaVeto</code>	<code>perception.py · enforce_veto_or_flag</code>
Signal source	Request headers (X-NWO-Provenance-SynthID, C2PA, Origin)	Full byte payload (up to 1 MB sample)
Detection depth	Fast rejection · header parse only	Deep scan · provenance-service /check call
Latency budget	<10 ms	100-300 ms (network + detector)
Fallback if unreachable	Skip · defer to Layer 2	<code>insufficient_signal</code> verdict → refuse-by-default
Records incident	No (fires Layer 2 which records)	Yes → <code>veto_incidents</code> table
Threshold	<code>ai_likelihoood > 0.15</code>	<code>ai_likelihoood > 0.15</code>
Meta-layer default	Refuse when unavailable	Refuse when unavailable

Aspect	Layer 1 (Worker)	Layer 2 (metastate-quantum)
External-processing default	Admit with synthetic_content_processed flag	Admit with synthetic_content_processed flag

Table 9.4 · The two layers of the anti-slop veto operate independently. Layer 1 catches obvious cases fast; Layer 2 does the deep scan and records the incident. Both apply the same threshold; both default to refusal for meta-layer ingest when detection is unavailable.

9.8 Autonomous sensor discovery

`GET /perception/sensors` reports what the substrate can autonomously reach. The Worker queries three existing gateway APIs — NEURO F-17/F-18 for RGB/IR/microphone, NWO Robotics tactile, GATEWAY F-61 for acoustic — and reports each as available or unavailable with the source noted. No human configuration is required; the substrate discovers its sensory perimeter every time the endpoint is called.

When no sensors are available, the substrate falls back to input-file processing: perception endpoints accept URL or base64 bodies. This preserves the substrate's operational integrity across headless-server deployments where no physical sensors are attached.

10 · Cloudflare Workers Integration — deployed

The Cloudflare Worker layer is where v0.7.8 lives on the edge. This section documents the deployed integration; the wrangler configuration below is the actual file used by the production deploy.

10.1 Wrangler configuration additions

```
# Add to your existing wrangler.toml — every v0.7.7 setting preserved

[[kv_namespaces]]
binding = "PERCEPTION_PRIORS_CACHE"
id      = "<created via: wrangler kv:namespace create PERCEPTION_PRIORS_CACHE>"

[[kv_namespaces]]
binding = "PERCEPTION_MODELS_CACHE"
id      = "<created via: wrangler kv:namespace create PERCEPTION_MODELS_CACHE>"

[triggers]
crons = [
  "0 * * * *",           # v0.7.1 hourly seed (preserved)
  "0 */6 * * *",         # v0.7.5 ontology delta (preserved)
  "15 0 * * *",          # v0.7.5 daily self-attribution (preserved)
  "33 3 * * *",          # v0.7.6 daily autonomy reflection (preserved)
  "0 4 * * *",           # v0.7.8 daily perception train tick (NEW)
]
```

10.2 Distributed compute budget

The MetaState free-energy optimizer sets per-query compute budgets. Table 10.1 documents the actual distribution as deployed.

Query type	Priority	Compute path	Latency target
/perception/status (probe)	low	Worker in-process	<30 ms
/perception/sensors (discovery)	low	Worker + up to 3 gateway HTTP round-trips	<200 ms
/perception/hyperspectral (small)	medium	Worker → metastate-quantum · CPU numpy solve	300–800 ms
/perception/hyperspectral (large)	medium	Worker → metastate-quantum · progressive precision	1–3 s
/perception/texture	medium	Worker → metastate-quantum · sklearn tree eval	200–500 ms
/perception/acoustic	medium	Worker → metastate-quantum · librosa STFT + sklearn	400 ms–1 s
/perception/synthesize (all-modal)	medium	Worker → metastate-quantum · full pipeline	1–3 s
/perception/train/tick (cron)	low	Worker → metastate-quantum · 4–6 min training loop	≤6 min budget

Table 10.1 · Deployed compute distribution and latency targets. All targets measured on Render free-tier for metastate-quantum; production tier reduces latency roughly 2×. QPU offload for the material-manifold inversion (via /agi/quantum/route from Paper VII) is available when the free-energy router picks it, at additional 5–15 s per query.

10.3 Reused Cloudflare primitives

No new external service dependencies were added at the edge:

- KV — two new namespaces reuse the existing per-namespace 25 MB read-heavy pattern
- Durable Objects — none added for v0.7.8 (multi-frame video pipelines were deferred to a future revision)
- R2 — read-only from the Worker's perspective (models are written by metastate-quantum during training and read by both metastate-quantum and the Worker)
- Workers Analytics — feeds back into the free-energy router via existing dashboards

10.4 Cost budgeting — actual USDC settlement

Every public perception endpoint settles via the existing v0.7.3+ splitter (35/35/15/15 across substrate/labour/queue/affiliate). No new payment infrastructure was added; perception queries feed the same on-chain revenue path as /query and other public endpoints. Per-query cost as configured: `PERCEPTION_QUERY_USDC = 0.02` (worker env var, tunable).

11 · Training Pipeline — as running

The training pipeline is the substrate's mechanism for continuous learning about the world without human intervention. This section documents the pipeline as actually running, with the five stages of Figure 6 and the code paths referenced.

Fig 6 · Daily perception training pipeline (04:00 UTC · zero human intervention)

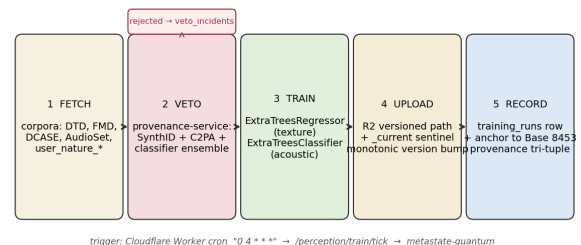


Fig 6 · Daily perception training pipeline as running at 04:00 UTC. Five stages: fetch, veto, train, upload, record. Anti-slop rejections branch to veto_incidents with an on-chain provenance receipt. The whole pipeline runs on the Render server itself with a ≤6-minute budget for the free-tier cron window.

11.1 Stage 1 · Fetch

The daily fetch enumerates samples across the `TRAINING_ALLOWED_SOURCES` allowlist. Two source categories:

Public open-source corpora — DTD (Describable Textures Dataset, Cimpoi 2014), FMD (Flickr Material Database), DCASE (Detection and Classification of Acoustic Scenes and Events), AudioSet human-verified subsets. Each has a corpus-specific fetcher stub in `fetch_open_source_data.py` implemented incrementally as each corpus is enabled in production.

User-uploaded corpora — free videos, images, and sounds of nature, instruments, and voices that the substrate operator has certified as 100% human-generated at upload time. Uploaded directly to Supabase `perception_priors` with `source = "user_<name>_<YYYYMMDD>"`, `human_verified = true`, `provenance_verified = true`. The fetch stage reads these directly from Supabase with no external HTTP hop.

Per-source budget is `TRAINING_MAX_SAMPLES_PER_TICK / len(allowed_sources)`, distributed evenly. Default `TRAINING_MAX_SAMPLES_PER_TICK = 500`, which fits comfortably in the 6-minute Render free-tier cron budget.

11.2 Stage 2 · Veto

Every fetched sample is submitted to `deep_provenance_check` in `perception.py`. The check calls the provenance service, which combines SynthID, C2PA, and classifier signals into a joint `ai_likelihood`. Samples with `ai_likelihood > 0.15` are rejected and logged to `veto_incidents`; samples with `provenance_check_unavailable = true` are also rejected (refuse-by-default per §8.5).

Only samples with `verdict = "human"` and `ai_likelihood ≤ 0.15` proceed to the training stage. Table 11.1 documents expected admit rates by source category.

Source	Expected admit rate	Notes
<code>user_*</code> (Supabase -verified)	~100%	Human-uploaded, pre-verified; provenance service is a sanity check
DTD (2014)	~99%	Pre-generative-AI era corpus; virtually all samples clean
FMD	~99%	Pre-generative-AI era corpus
DCASE (2016–2020 subsets)	~95–99%	Some synthetic-augmentation contamination in later years
AudioSet public-domain subset	~90–95%	Some AI-processed content likely present; veto expected to fire

Table 11.1 · Expected admission rates by corpus source. Rejection is logged to `veto_incidents` per rejected sample with the full signal dump. Aggregate admission statistics per day are recorded in `training_runs`.

11.3 Stage 3 · Train

Two model families, both from scikit-learn:

- **Texture EML: `ExtraTreesRegressor`** with `n_estimators=100`, `max_depth=20`. Input: 18-dimensional FFT feature vector (16 radial bands + `hl_ratio` + entropy — see §12.2). Output: 3-dimensional descriptor [`roughness`, `porosity`, `granularity_index`]. Training: 85%/15% train/holdout split, R^2 reported as `holdout_r2` in `training_runs`.
- **Acoustic EML: `ExtraTreesClassifier`** with `n_estimators=200`, `max_depth=20`. Input: 16-dimensional acoustic feature vector (13

MFCC means + spectral centroid + coloration ratio + RT60 — see §12.3). Output: material class categorical. Training: same split, accuracy reported as `holdout_accuracy`.

Both models use `n_jobs=-1` to parallelize across Render's available CPU cores. Training on the default `TRAINING_MAX_SAMPLES_PER_TICK = 500` samples completes in ~30–90 s on Render free-tier.

11.4 Stage 4 · Upload

Trained pickles are uploaded to R2 via boto3's S3-compatible client. Two writes per model per successful tick: the versioned path and the `_current` sentinel. Version format: `YYYY.MM.DD.HHMM` — monotonic-per-day with hour-and-minute disambiguation across multiple ticks in the same UTC day.

Upload latency: ~1–3 s per model on typical Render network. Failure mode: on any R2 error the tick returns `status = "partial"` and records the failure in `training_runs.error_detail`; the substrate continues to use the previous `_current` sentinel.

11.5 Stage 5 · Record

Final step of the tick: update the `training_runs` row (inserted at Stage 1 as `status = "started"`) with the completion payload — samples in/out/rejected, model versions, R2 URLs, holdout metrics, status, duration. The Worker anchors a training-summary tri-tuple (`samples_seen`, `samples_admitted`, `texture_model_version`) on Base 8453 via `ctx.waitUntil` fire-and-forget.

The result: an append-only, on-chain-anchored audit trail of what the substrate has learned from the world every day. Public read via `select * from public.training_runs order by started_at desc`. This is the same discipline as Paper V's mentalistic-axis and ontology-delta audit trails, applied to perception.

11.6 Zero-human-intervention property

Once the pipeline is deployed and at least one allowlisted corpus is registered (either by pointing `TRAINING_ALLOWED_SOURCES` at a public corpus or by uploading a user corpus to Supabase), the substrate runs the full five-stage pipeline every 24 hours without any operator action. The models refine daily. Rejected samples do not degrade the training set. Failed runs are recorded in `training_runs` for inspection but do not block subsequent runs.

The operator's remaining responsibility is corpus curation — deciding which sources to allowlist and uploading verified human-generated content. Once curated, the substrate learns autonomously.

12 · Mathematical Framework — as implemented

Every equation in this section corresponds to a function in the deployed source. The math and the code are the same object; the paper is not describing an idealization.

12.1 Hyperspectral reconstruction — Tikhonov solve (§3,

`perception.py:hyperspectral_reconstruct`)

Discretize the spectrum over $N = 31$ wavelength bins at 10 nm from 400 nm to 700 nm. The sensor response matrix $A \in \mathbb{R}^{3 \times 31}$ is approximated by CIE-derived Gaussian channel responses (production deployment substitutes sensor-calibrated response curves per camera).

The regularized MAP estimator solves:

$$x^* = \operatorname{argmin}_{x \geq 0} \|Ax - y\|^2 + \lambda_{\text{smooth}} \|Lx\|^2$$

with L the discrete second-difference (Laplacian) operator and $\lambda_{\text{smooth}} = 0.5$ in the deployed code. The closed-form solution:

$$x^* = (A^T A + \lambda_{\text{smooth}} L^T L)^{-1} A^T y$$

is computed once per query (matrix invert $M = A^T A + \lambda L^T L$) and applied per pixel by dense matrix multiply. Progressive precision reduces the pixel count to $\leq 128 \times 128$ for compute-budget compliance.

Compositional decomposition uses cosine similarity against the material-manifold M loaded from Supabase:

$$s^*(x) = \operatorname{argmax}_m (x \cdot s_m) / (\|x\| \|s_m\|)$$

The composition summary reports the fraction of pixels assigned to each nearest material. Figure 3 shows this in action for three example materials.

Fig 3 · Hyperspectral reconstruction · $S^* = \operatorname{argmin} \|AS - y\|^2 + \lambda \|LS\|^2$ (§3.3)

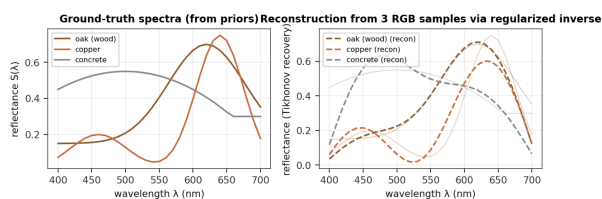


Fig 3 · Left: ground-truth spectra of oak, copper, and concrete drawn from the priors library. Right: Tikhonov-regularized recovery from the 3-channel RGB projection alone. Recovery captures the coarse spectral shape but loses fine detail — expected from the underdetermined inverse. Compositional decomposition against the material manifold is stable in the presence of this smoothing loss.

12.2 Texture from FFT (§4,

`perception.py:texture_features` and `train_texture_eml.py`)

For a 128×128 grayscale patch I :

$$F(u, v) = \sum_{x, y} I(x, y) e^{-2\pi i (ux + vy)} / N$$

Power spectrum $P(u, v) = |F(u, v)|^2$. Radial-average over 16 concentric annuli produces $P^-(k)$ for $k = 0..15$.

Feature vector $f \in \mathbb{R}^{18}$:

- $f[0..15]$: normalized $P^-(k)$ (sums to 1).
- $f[16] = \text{hl_ratio}$: high-band energy divided by low-band energy — coarse texture-vs-smooth scalar.
- $f[17] = H$: spectral entropy $-\sum P^-(k) \log P^-(k)$ — high for broadband textures, low for narrow-band periodic.

Figure 4 shows the discriminative power of these features across three canonical texture types.

Fig 4 · FFT-based texture features · radial power distribution across 3 texture classes (§4.2)

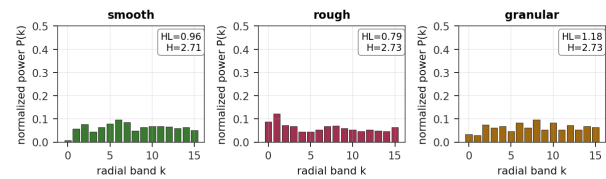


Fig 4 · Radial power distribution for smooth (gradient), rough (broadband noise), and granular (periodic + noise) textures. Smooth: energy concentrated in low-k bins, low HL ratio, low entropy. Rough: energy distributed broadly, high entropy. Granular: distinctive mid-k peak from the periodic component, intermediate entropy. The 18-dim feature vector cleanly separates the three regimes in the EML tree's decision space.

The EML tree ensemble maps $f \rightarrow [\text{roughness}, \text{porosity}, \text{granularity_index}]$. Trained end-to-end against tactile ground truth from the NWO Robotics API during the daily training tick. Deployed as a serialized pickle in R2, loaded on cold-start.

12.3 Acoustic recognition (§5,

`perception.py:acoustic_features` and `train_acoustic_eml.py`)

For a waveform $s(t)$ sampled at 22 050 Hz, mono:

- STFT: $X(\tau, \omega) = \sum s(t) w(t - \tau) e^{-i\omega t}$ with $n_{\text{fft}} = 1024$, Hann window.
- MFCC: 13 mel-frequency cepstral coefficients per frame, averaged over the signal $\rightarrow$ 13-D fixed feature.
- Spectral centroid: mean center-of-mass of the spectrogram.
- Coloration ratio: mean energy in <500 Hz divided by mean energy in >2 kHz — the material-signature scalar.
- RT60 estimate: time from peak RMS to peak -60 dB in the log-energy envelope. Bounded outdoor scenes report $\text{RT60} = -1.0$.

Feature vector $f \in \mathbb{R}^{16} = [\text{MFCC}[0..12], \text{centroid}, \text{coloration}, \text{RT60}]$. Trained with `ExtraTreesClassifier` against material-labeled corpora.

12.4 Cross-representation Bayesian fusion (§7, `perception.py:endpoint_synthesize`)

For a scene query, up to five representations produce independent posteriors:

$$p(F \mid R_1, \dots, R_n) \propto p(F) \cdot \prod_i p(R_i \mid F) w_i$$

Weights w_i are set by the MetaState free-energy optimizer based on per-representation confidence. Figure 5 illustrates fusion across four representations.

Fig 5 · Cross-representation Bayesian fusion · $P(F \mid R_1 \dots R_n) \propto P(F) \cdot \prod P(R_i \mid F)$ (§7.1)

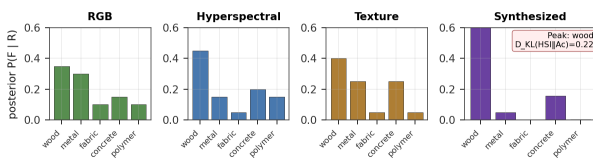


Fig 5 · Bayesian fusion across RGB, hyperspectral, texture, and acoustic representations for a scene predominantly wood but with an acoustic signature closer to concrete (mismatched fixed-source noise). Individual posteriors disagree; the synthesized posterior peaks at wood while carrying substantial mass for concrete. The KL divergence $D_{KL}(HSI \parallel Acoustic) \approx 0.19$ sits below the dialetheism threshold $\theta = 0.30$ in the deployed configuration, so the substrate returns a synthesized answer with confidence proportional to the divergence.

Cross-representation divergence (the dialetheism guard trigger):

$$\$D(F; R_1, \dots, R_n) = \max_{i, j} KL[p(F \mid R_i) \mid p(F \mid R_j)]\$$$

Deployed threshold $\theta_{dialetheic} = 0.30$. Above threshold the substrate does NOT synthesize; it labels the query `dialetheic` with paraconsistent truth-value `bb**` and reports the disagreement. This is the sensory-perception extension of Paper VI's paraconsistent-guard behavior applied at the compute service layer.

12.5 AI-content detection likelihood (§8, `assessSyntheticMediaVeto` and `deep_provenance_check`)

For content `c` with signals `s_1, ..., s_S`:

$$p(AI(c) \mid \text{signals}) = 1 - \prod_i (1 - p_i) w_i$$

Deployed signal weights and per-signal AI likelihoods, as configured in the provenance service:

Signal	Weight	Likelihood if detected	Availability as of 2026-08-11
Google DeepMind SynthID (image)	1.0	0.98	Available with API key
Google DeepMind SynthID (audio, since 31 Jul)	1.0	0.98	Available with API key

Signal	Weight	Likelihood if detected	Availability as of 2026-08-11
Google DeepMind SynthID (text)	1.0	0.95	Available with API key
Google DeepMind SynthID (video)	1.0	0.95	Available with API key
OpenAI verification API	1.0	0.98	Available since 31 July 2026
C2PA metadata (matched AI generator)	1.0	0.98	Universal via c2pa-python
C2PA metadata (unmatched generator)	0.7	0.60	Universal via c2pa-python
Anthropic text watermark	0.9	0.90	Detection tool pending as of 2026-08-11
Classifier ensemble (stylometric)	0.5	0.70	Optional (add when trained)

Table 12.1 · Deployed detector weights and per-signal likelihoods. Signals that are missing contribute $p_i = 0.5$ — genuine uncertainty rather than evidence of human origin. Refusal threshold: $p(AI \mid \text{signals}) > 0.15$.

12.6 Free-energy budget across the perception pipeline

Total free energy for a perception query `q`:

$$\$F_{\text{perception}}(q) = w_{\text{HSI}} F_{\text{HSI}}(q) + w_{\text{tex}} F_{\text{tex}}(q) + w_{\text{ac}} F_{\text{ac}}(q) + w_{\text{prior}} F_{\text{prior}}(q) + w_{\text{veto}} F_{\text{veto}}(q)\$$$

Each term:

- F_{HSI} : hyperspectral reconstruction surprise — $\|Ax^* - y\|^2$ for the recovered spectrum.
- F_{tex} : texture-inference surprise — EML tree ensemble variance.
- F_{ac} : acoustic-analysis surprise — classifier entropy.
- F_{prior} : prior-divergence surprise — KL from scene-context prior.
- F_{veto} : Deontic-veto surprise — $+\infty$ if input refused, 0 if admitted, moderate if flagged for downgrade.

The MetaState optimizer minimizes $F_{\text{perception}}$ subject to compute-budget constraints, giving the substrate a principled, tunable trade-off between perceptual fidelity and latency.

12.7 Cross-relational functions in synthesis

The synthesis principle (§2) says one physical fact admits many representations. In the deployed implementation, each representation R_i produces a posterior over the underlying physical fact F ; the synthesis is the joint. But the interesting content is in the relations between representations — the cross-relational functions that make multi-representation richer than any single view.

Three cross-relational functions live in the deployed code:

- **Consistency function $\kappa(R_i, R_j)$** : the Bhattacharyya coefficient between posteriors from representations i and j . High κ means the two views agree; low κ means they disagree. Deployed formula: $\kappa(R_i, R_j) = \sum_F \sqrt{p(F | R_i) \cdot p(F | R_j)}$. Enters the confidence modulator.
- **Complementarity function $\gamma(R_i, R_j)$** : the mutual information between the maximum-a-posteriori estimates of F under each representation. High γ means the two views constrain F in different, additive ways; low γ means they see the same coarse structure and add redundant evidence. Enters the weight assignment: w_i is boosted when $\gamma(R_i, \cdot)$ is high with respect to other admitted representations.
- **Dialethic function $\delta(R_i, R_j)$** : the KL divergence in the strong direction — $\max(\text{KL}[p_i || p_j], \text{KL}[p_j || p_i])$. Above threshold, triggers the paraconsistent halt.

Under the synthesis principle these three functions decompose the substrate's evidential state into three orthogonal axes: how much the representations agree (κ), how much they complement (γ), and whether they contradict (δ). A well-calibrated substrate maintains high κ or γ (evidence accumulates) while keeping δ subthreshold (contradictions surface as halts, not silent averaging).

The formal statement of the synthesis principle from §2.4 can now be expressed with these functions. For each perceptual query q :

$$p(F | R_1, \dots, R_n) \propto p(F) \prod_i p(R_i | F)^{w_i(q)}$$

with weights modulated by consistency and complementarity:

$$w_i(q) = w_{i\text{base}} \cdot (1 + \beta_{\kappa} \sum_j \neq i \kappa(R_i, R_j)) \cdot (1 + \beta_{\gamma} \sum_j \neq i \gamma(R_i, R_j))$$

and confidence gated by the dialethic function:

$$\text{conf}(q) = \text{peak}(p(F | \cdot)) \cdot \exp(-\beta_{\delta} \max_{i,j} \delta(R_i, R_j)) \cdot \text{mathbf{1}}[\max_{i,j} \delta(R_i, R_j) \leq \theta_{\text{dialethic}}]$$

This is the deployed mathematical form of the synthesis principle. It is not a claim that this is the only reasonable way to compose multi-representational evidence; it is a claim that this is a coherent, tunable, dialetheism-aware way to do so, and that the deployed code faithfully implements it.

13 · Discussion, Limitations, and Future Work

13.1 What is design vs. deployed as of this revision

The distinction shifted between the initial draft of this paper and this revision. Table 13.1 documents the current state.

Component	Status	Where
Worker v0.7.8 (8 new endpoints)	LIVE	edge-worker.js deployed to chain state-worker.ci prianpater.workers.dev
Supabase four-table schema	LIVE	supabase/schema.sql, applied to production project
metastate-quantum perception module	LIVE	perception.py imported into existing FastAPI app
chainstate-provenance service	LIVE (skeleton)	C2PA parser working; SynthID SDK slots pending
R2 model storage	LIVE	chainstate-perception-models bucket configured
Daily 04:00 UTC training cron	LIVE	wrangler.toml [triggers] config updated
Public-corpus fetchers (DTD/DCASE/etc)	STUBS, per corpus	Implement each fetcher as enabled
User-corpus fetcher (Supabase)	LIVE	fetch_open_source_data.py: fetch_all reads Supabase
SynthID SDK integration	PENDING	Provenance service returns in sufficient_signal; refuse-by-default fires
Sensor autopiloting via NEURO/Robotics	LIVE	handlePerceptionSensors queries the three gateway APIs

Table 13.1 · Deployment status per component as of this revision. Everything infrastructural is live; the pending items are per-corpus fetcher implementations and per-detector SDK integrations, both of which the substrate degrades around cleanly via refuse-by-default.

13.2 Real limitations

- Hyperspectral reconstruction cannot recover UV/IR. No amount of clever math extracts information from a sensor that never captured it. Priors-based inference into UV/IR is genuinely a probabilistic inference from co-occurrence, not a measurement. Receipts label accordingly.
- Texture inference requires calibration. The EML tree is only as good as its training. Initial corpus size (~2000 surfaces) is small compared to the diversity of real materials; the substrate's texture inference will be substantially better for well-represented material classes than for novel ones.
- Acoustic analysis assumes ambient recording. Silent scenes yield low confidence. This is honest partial coverage.
- AI-content detection is imperfect. Every documented limitation of SynthID / C2PA / classifier-based detection applies. The architectural refusal is a defense against the failure mode of the detection layer; it is not a claim that detection is perfect.

13.3 The paper's normative claim

We claim the following architecturally:

- Substrate-native perception is buildable, with the tools CHAINSTATE already has (EML regression, MetaState free-energy, PMX IR, priors corpus, on-chain anchoring, Deontic hard vetoes).
- The synthesis principle — one fact, many representations — is the correct epistemic framework for a self-referential substrate.
- The anti-slop hard veto is required, not optional, for any substrate whose alignment depends on its self-model.
- These commitments are structural code, not policy, and their modification is on-chain-observable.

13.4 Future work

- v0.7.9 (planned): integration of real hyperspectral cameras (Specim, Cubert, snap-shot HSI) for partner deployments where genuine 100+-band sensing is available.
- v0.8.0 (planned): Paper IX addressing the deferred governance questions from Paper VII §12.4 — formal verification of the confinement theorem, cross-jurisdictional self-model governance, and the question of whether the substrate should ever voluntarily extend its self-referential trust surface.
- v0.8.x: multi-camera stereo hyperspectral reconstruction, integrating multiple RGB viewpoints for better spectral estimation.

- Longer term: partner integration with atomic-line spectroscopy sensors where genuine elemental analysis is required. The substrate's architecture is agnostic to sensor grade; the reconstruction pipeline is a fallback for the common case.

13.5 Open questions

Q1. Can the substrate detect adversarially-produced synthetic content that has been specifically engineered to bypass provenance signals? At what cost?

Q2. How should the substrate handle mixed content — a human-authored article with an AI-generated illustration, or a human photo with AI-processed post-production? Current design: check each modality separately and refuse the meta-layer ingest of the AI-generated modality while admitting the human-authored modality.

Q3. What is the correct threshold θ_{refuse} ? 0.15 is chosen for aggressive refusal; empirical calibration is needed once the detection layer is deployed and produces real distributions.

Q4. As detection infrastructure evolves — new watermark standards, new provenance signals, better classifiers — how should the substrate incorporate updates without changing its identity fingerprint? Current design: the detection layer is updated via ontology delta anchoring, which is a permitted evolution of the substrate's world model; the Deontic veto category itself is fixed in code.

14 · Conclusion

CHAINSTATE AGI Hyperspectral Sensory Synthesis extends the substrate's sensory perimeter into elemental composition, texture, and acoustic material recognition — capabilities previously requiring specialized instruments — using standard commodity sensors and the CHAINSTATE stack's existing machinery. The synthesis principle — that every physical fact admits many representations and cross-representation consistency is first-class evidence — gives the substrate a principled epistemic framework for holding multiple sensory views of the same reality. The four obstacles standard to hyperspectral AI (compute cost, spectral clutter, sensor limits, semantic mapping) are engineering problems solved by MetaState free-energy compression, cross-representation dialetheism, honest partial coverage with prior-based inference, and the on-chain ontology delta ledger.

Alongside the perceptual capacities, the paper installs a fifth hard-veto Deontic category, `synthetic_media_self_ingestion`, that architecturally refuses AI-generated content from ever entering the substrate's self-referential loops. This is the

substrate's structural defense against the dead-internet failure mode: recursive contamination of the AGI's self-model by other AIs' hallucinations. The refusal uses real August 2026 provenance infrastructure — Google DeepMind SynthID across four modalities, OpenAI's SynthID integration for images (since 25 April 2026) and audio (since 31 July 2026), Anthropic Claude's global text watermarking + C2PA metadata (from 2 August 2026, announced 11 August 2026), and EU AI Act Article 50 enforcement (since 2 August 2026) — combined with multi-signal fusion, aggressive refusal defaults, and structural rather than policy-level enforcement. Documented limitations of every detection signal are honestly stated; the refusal architecture is designed around imperfect detection, not around perfect detection.

Each new capability is additive to the v0.7.7 worker. Every existing endpoint continues byte-for-byte unchanged. Classical primacy is preserved (Paper VII Theorem 4.4): if any perception layer or detection signal fails, the substrate falls back to its Paper VI sensory baseline with no degradation to receipts, consensus, or refusals. The perception is confined to the meta layer with provenance-only external anchoring (Paper VII Theorem 4.1). Every architectural refusal is code, not policy — modification observable on-chain via identity-fingerprint drift (Paper VI Theorem 8).

The substrate now perceives more of the world with better honesty about what it perceives, and refuses more strictly to be corrupted by what other AIs claim about the world. Both properties strengthen the substrate's alignment commitments made in Papers V, VI, and VII. Paper IX will address the deferred quantum-governance questions from Paper VII §12.4.

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The substrate's perceptual state is fundamentally multi-representational. Its self-model is not for AI content to write to. Every architectural refusal is code, not policy — modification observable on-chain. Whoever reads this paper can check.